

Have a more beautiful life

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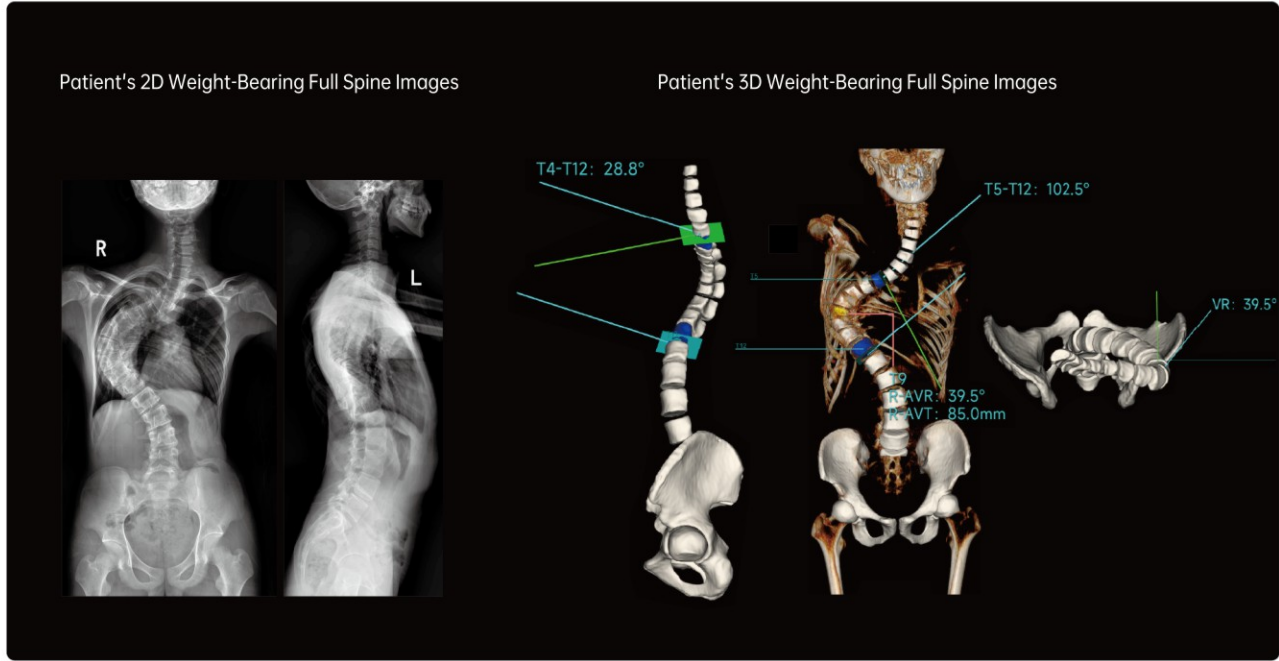
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WeR-3D *Weight-Bearing
Radiographic 3D System*

Clinical Application Case: Full Spine Imaging

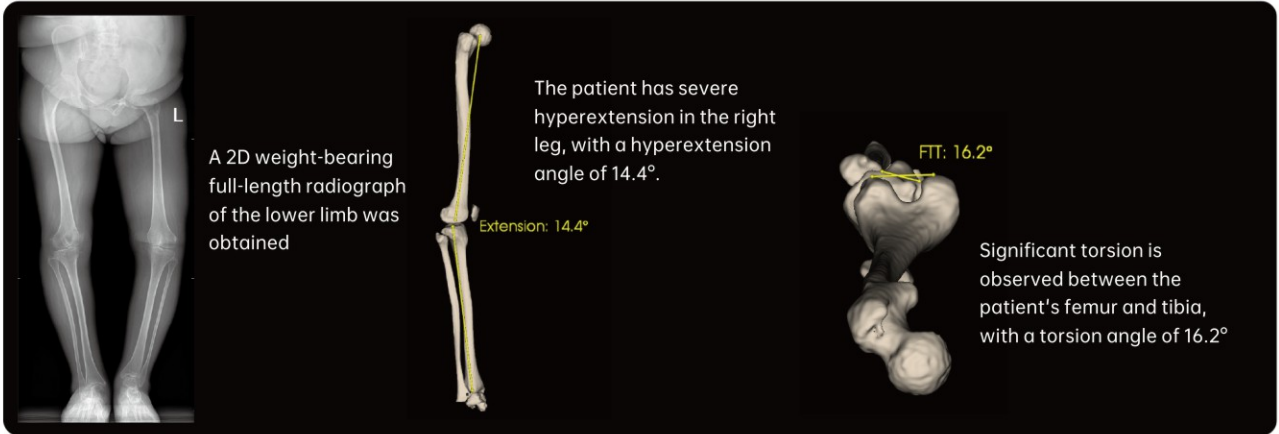


Measured Parameters	2D Measurement	3D Measurement	Absolute Difference Cobb
Cobb	102.1°	102.5°	0.4°
Apical Vertebral Rotation	Unable to measure	39.5°	/
T4-T12 Kyphosis Angle	Difficult in Severe Cases	28.8°	/
L1-L5 Lordosis Angle	57.3°	43.9°	13.4°
Affected by Pelvic Tilt	12mm	9mm	3mm

Traditional 2D imaging is prone to projection-induced errors and offers limited parameter options, failing to accurately quantify vertebral rotation. It struggles with severe deformities and relies heavily on manual interpretation, which often underestimates the true severity of scoliosis.

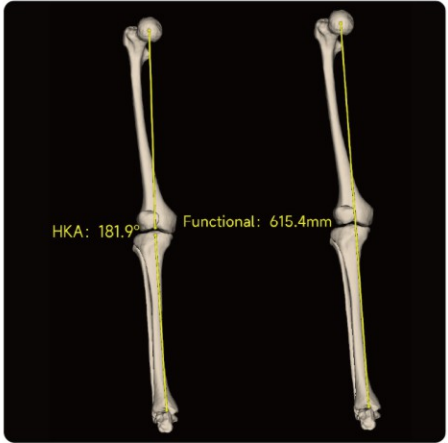
In contrast, 3D imaging faithfully reconstructs the spinal anatomy in three dimensions, enabling precise measurement of vertebral rotation, Cobb angles, and other key parameters. This significantly improves the accuracy of scoliosis classification, progression monitoring, and treatment planning.

Clinical Application Case: Full Lower Limb Imaging



Right Leg	2D Measurement	3D Measurement	Absolute Difference
Functional Length	630mm	615mm	15mm
Tibial Length	294.4mm	286.8mm	7.6mm
Femoral Length	330.5mm	321.6mm	8.9mm

In cases of severe sagittal plane deformity, conventional 2D full-length radiographs suffer from projection inaccuracies due to variations in depth among bony structures. Regions of interest lie at different distances from the detector, leading to magnification errors and unreliable length measurements.



Right Leg	2D Measurement	3D Measurement	Absolute Difference
MAD	-31.7mm	-17.2mm	14.5mm
HKA	191.7°	181.9°	9.8°
mMTPA	89.5°	81.6°	7.9°
mLDFA	90.1°	81.6°	8.5°

In cases of severe sagittal plane deformity, conventional 2D full-length radiographs suffer from projection inaccuracies due to variations in depth among bony structures. Regions of interest lie at different distances from the detector, leading to magnification errors and unreliable length measurements.

WR-3D: Weight-Bearing Radiographic System+

The WR-3D system utilizes a cone-beam CT imaging technique. During operation, the X-ray tube and detector remain stationary while the motorized platform rotates the patient through a full 360-degree cycle at a uniform speed. Proprietary algorithms are then employed to reconstruct weight-bearing three-dimensional images. The system currently supports weight-bearing 3D scanning of the full spine and full lower extremities, offering a maximum FOV of 1310 mm in the coronal plane and 350 mm in the axial plane.



Maximum axial FOV: 350 mm
The WR-3D system employs a non-isocentric scanning design, supporting full abdominal and full lower extremity imaging.



Maximum coronal FOV: 1310 mm
It enables 3D scanning of the full spine and full lower limbs.



Low-dose weight-bearing 3D imaging
With a radiation dose equivalent to only 10%–20% of typical CT protocols, the system accurately captures joint alignment and biomechanical status under loaded conditions.

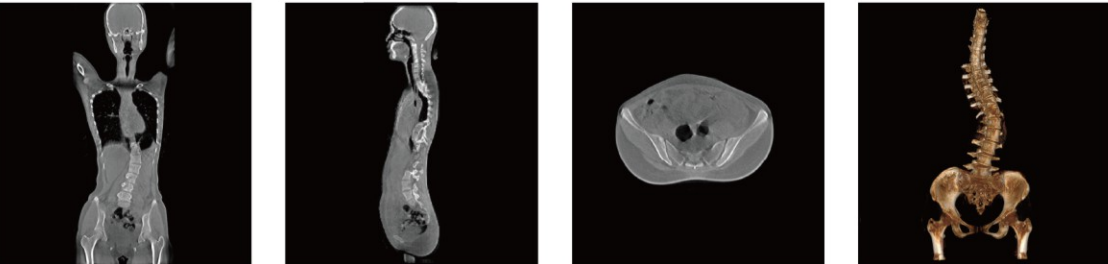


It provides precise visualization of sagittal alignment and complex spatial rotations of bones and joints in patients.



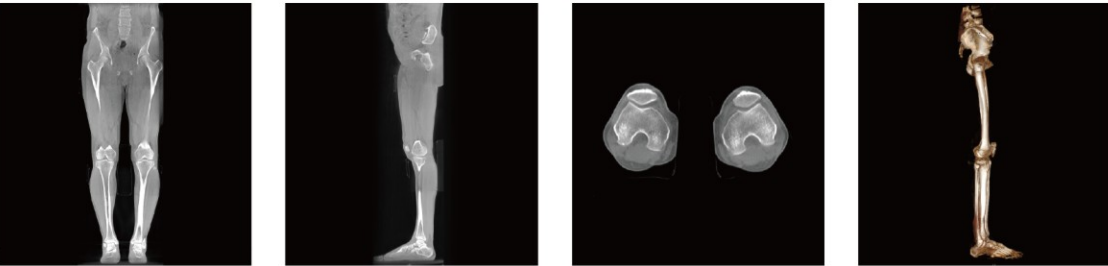
Full Spine 3D Imaging

WR-3D facilitates weight-bearing 3D scanning of the entire spine. Compared to traditional 2D or recumbent 3D modalities, it delivers more accurate and comprehensive imaging data, supporting precise diagnosis, surgical planning, and post-operative evaluation for conditions such as scoliosis.



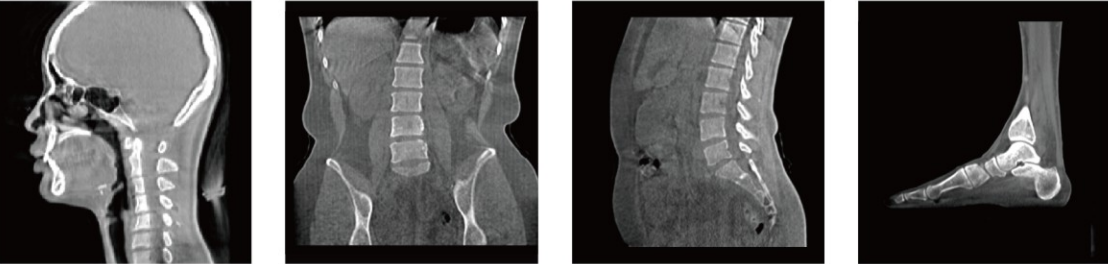
Full Lower Limb 3D Imaging

The system offers weight-bearing 3D imaging of the full lower extremities. It provides enhanced assessment of lower limb alignment and biomechanical parameters over conventional 2D or non-weight-bearing 3D techniques, delivering valuable clinical evidence for bracing, orthopedic corrections, and surgical decisions in cases like knee osteoarthritis.



Whole Body Part 3D Imaging

WR-3D enables weight-bearing 3D scanning and analysis of multiple anatomical parts, including the cervical spine, lumbar spine, ankle, knee joint and airway. A single-part 3D scan is completed in just 25 seconds, enabling efficient and focused diagnostic applications.

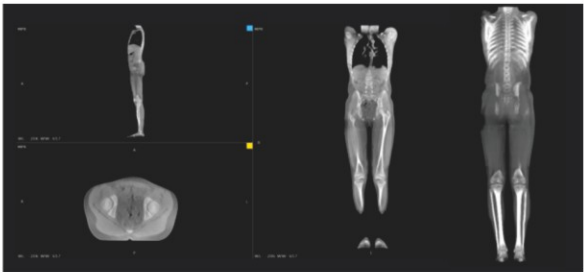


World's First Weight-Bearing Full Spine Scanning and Measurement System⁺

WR-3D delivers intuitive and precise quantitative data for orthopedic scoliosis assessment, offering vital clinical support throughout surgical planning and follow-up evaluation.

Weight-Bearing 3D Scanning & Reconstruction: Full Spine, Full Lower Limbs, and Whole Body

Using an advanced multi-sequence scanning method based on cone-beam CT technology, WR-3D achieves true weight-bearing 3D imaging of the full spine, full lower limbs, and whole-body parts.



MIP – Maximum Intensity Projection

MIP clearly displays the overall structure of the patient's spine, supporting high-contrast visualization of bony anatomy.



MPR – Multi-Planar Reformation

MPR enables interactive multi-plane diagnosis and measurement—including axial, sagittal, and coronal views—unlocking 3D spatial insights not possible with 2D imaging.



VR – Volume Rendering

With advanced computer rendering, WR-3D generates detailed 3D volume reconstructions, helping doctors visualize anatomy more intuitively.

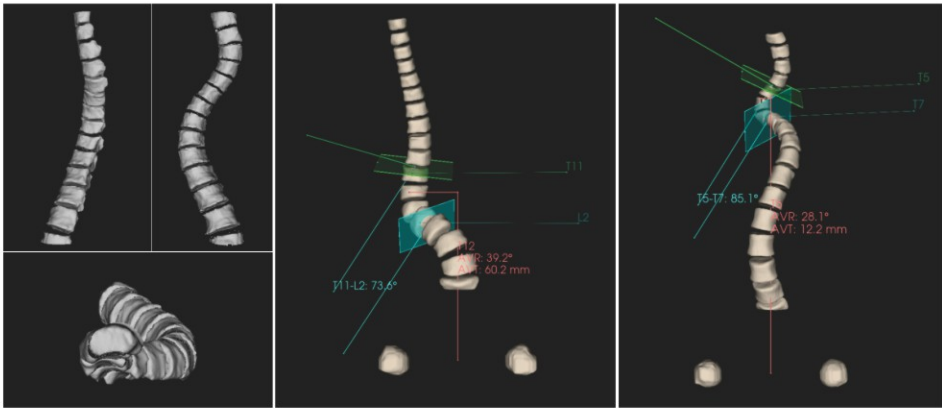


AI-Powered Vertebra Recognition and Labeling

Incorporating AI-based image processing, the system automatically identifies, segments, and measures key spinal parameters. Current functions include 3D Cobb angle measurement, vertebral rotation analysis, and comprehensive scoliosis evaluation—streamlining diagnosis, planning, and post-operative assessment.

Multi-Plane Scoliosis Quantification

WR-3D supports accurate quantitative evaluation in both coronal and sagittal planes—including cervical lordosis, kyphosis, and lumbar lordosis—enabling complete deformity assessment.



World's first fully automatic AI measurement system⁺

WR-3D is the world's first system capable of fully automated weight-bearing full-spine and full-lower-limb measurements. It delivers intuitive and highly accurate quantitative data for orthopedic applications such as scoliosis and knee osteoarthritis assessment, offering essential clinical value in preoperative planning and postoperative evaluation for spinal and lower limb orthopedic surgeries.

Moreover, WR-3D enables one-click generation of measurement reports in just 15 seconds—60 times faster than the average orthopedic measurement time of 15 minutes. This significantly enhances clinical workflow and supports clearer, evidence-based doctor-patient communication by visually demonstrating diagnostic and therapeutic outcomes.

The system also supports 3D model output of scanned body parts for use in orthopedic brace design, surgical planning, and medical training.



Delivers weight-bearing sagittal alignment and 3D rotational data



Employs limb-specific coordinate systems to reflect true anatomical deformity



Improves measurement efficiency by 60x

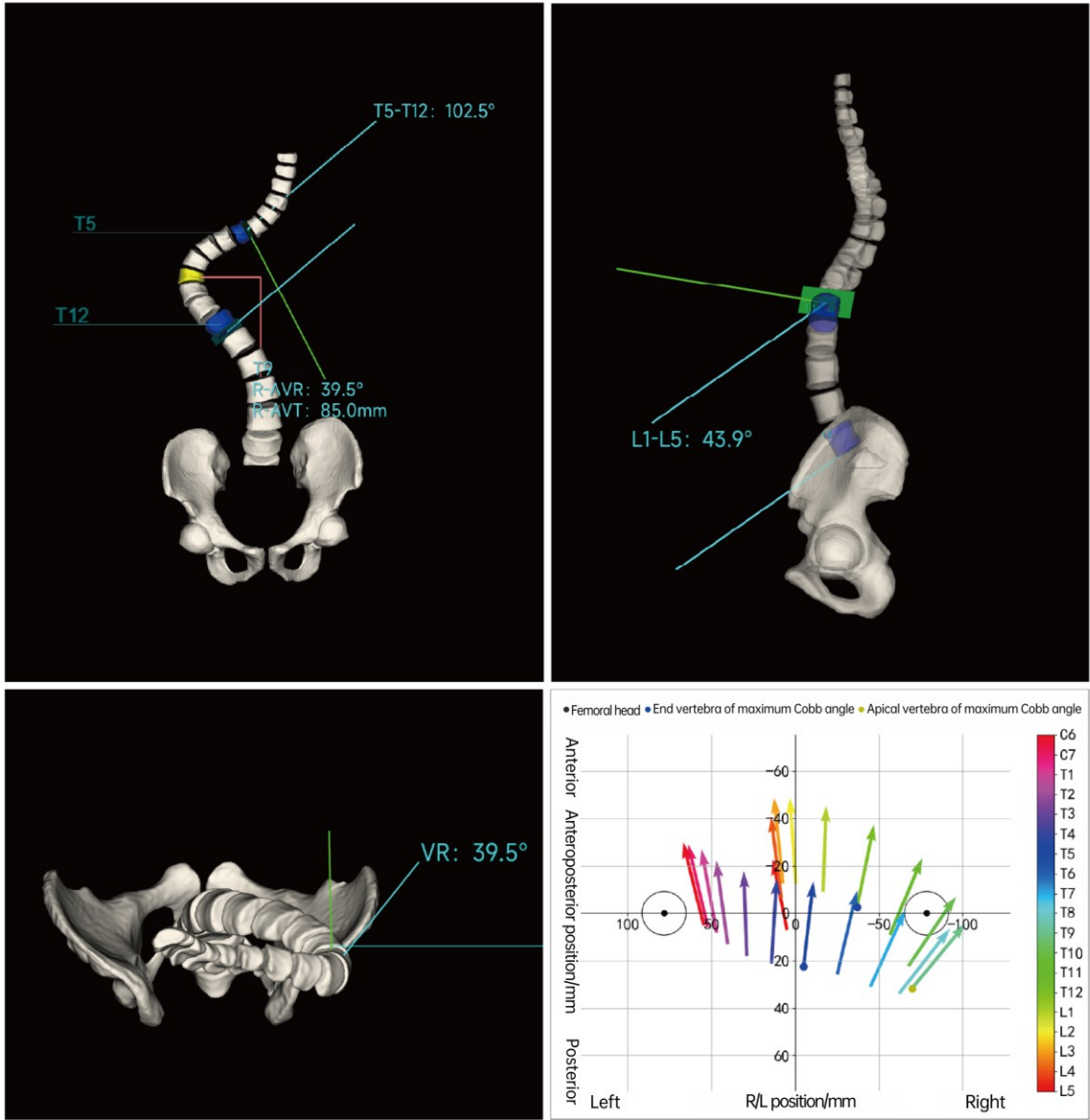


Full Spine Measurement System

WR-3D performs comprehensive spinal and pelvic measurements, including automated analysis of:

- 3D Cobb angles
- 3D vertebral rotational angles
- Pelvic tilt
- Sagittal vertical axis (SVA)

It also supports automated outputs of 3D printable files of the full spine (both skeletal and surface models).



World's first fully automatic AI measurement system⁺

Full Lower Limb Measurement System

WR-3D performs fully automated measurements of the entire lower extremity, supporting a comprehensive set of parameters including:

- Functional length
- Femoral and tibial torsion angles
- Mechanical Axis Deviation (MAD)
- Hip-Knee-Ankle Angle (HKA)
- Joint Line Convergence Angle (JLCA)
- mL DFA (mechanical Lateral Distal Femoral Angle)

It also automatically outputs 3D printable documents (both skeletal and surface) of the full lower limb.

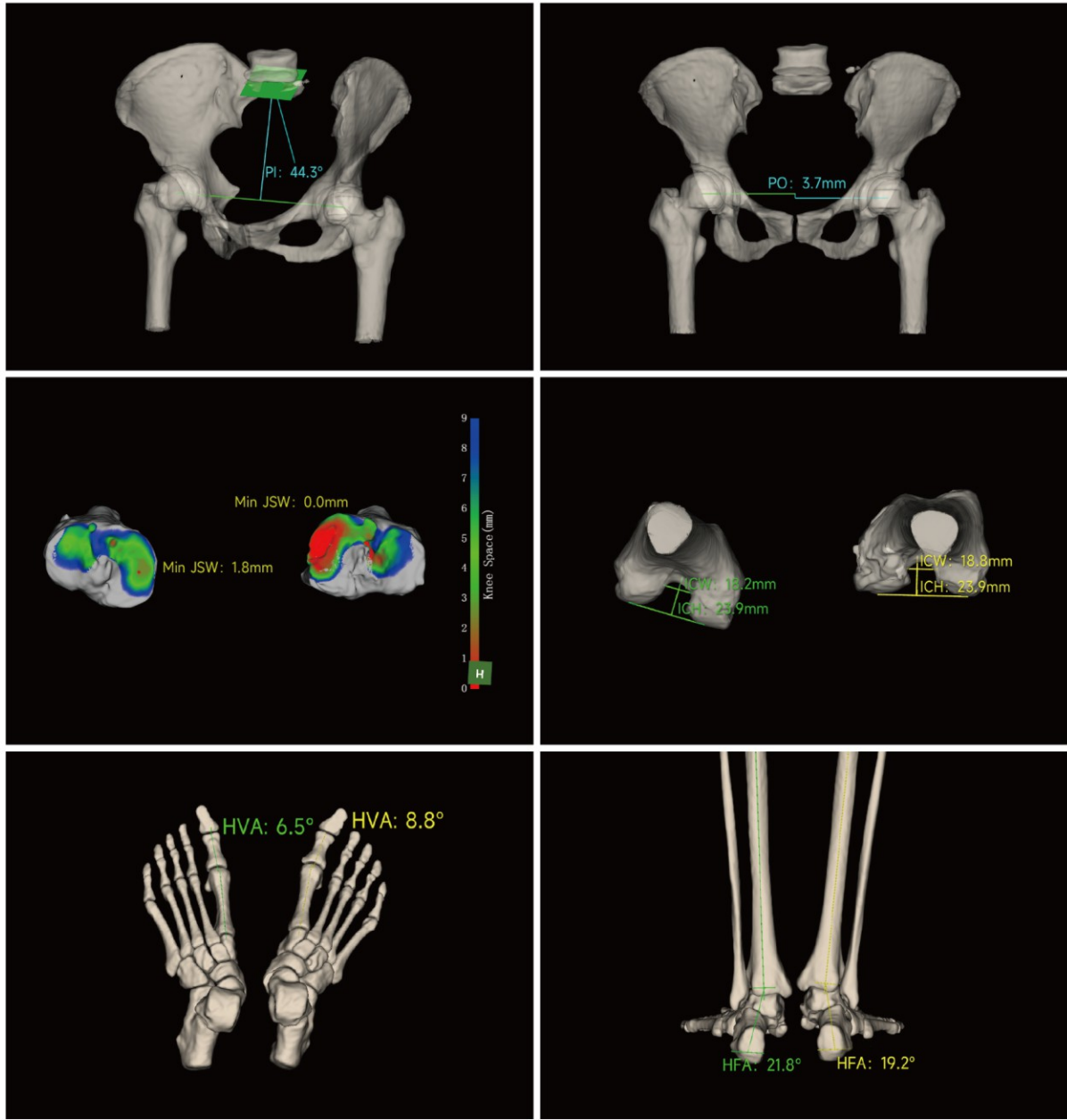


Single-Part Measurement System

WR-3D enables automated analysis of specific joint regions—ankle/foot, knee, and hip—with support for key parameters such as:

- Meary's angle
- Hallux valgus angle
- Knee convergence angle
- Joint space width
- Pelvic incidence

It also automatically outputs 3D printable documents for each specific anatomical region.



3D+2D scanning system⁺

WR-3D supports integrated 2D and 3D imaging protocols. In addition to weight-bearing 3D scans, it performs conventional X-ray imaging, full-spine and lower-limb image stitching, contrast-enhanced examinations, fluoroscopy, and pneumoconiosis exams—meeting diverse clinical needs within a single system.



High Operational Efficiency

Without altering existing device workflows, WR-3D enables both 2D and 3D examinations within the same room. This eliminates the need for patient transfer between systems, significantly improving throughput and streamlining operations.



Maximized Room Utilization

A single system supports:

- 3D scanning of single parts, full spine, and full lower limbs
- 2D examinations including radiography, fluoroscopy, contrast studies, pneumoconiosis exams, and full-length stitching

This versatility allows diversified services within one room, maximizing space efficiency and enhancing departmental operational performance.

Full application



Contrast Examination



Conventional X-ray



Full-Spine/Lower-Limb Image Stitching

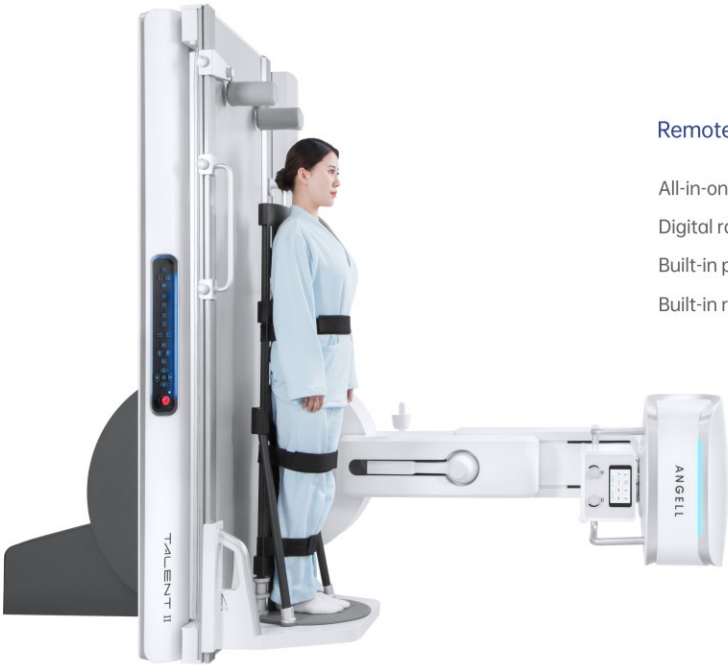


Fluoroscopy



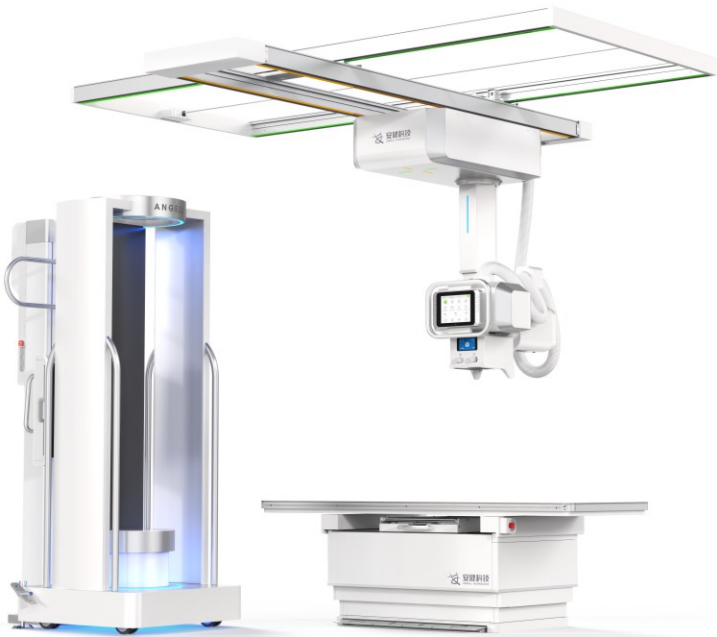
Pneumoconiosis Examination

Other models with WR-3D⁺



Remote-Control-Table Dynamic DR

- All-in-one WR-3D imaging solution
- Digital radiography and digital fluoroscopy
- Built-in pneumoconiosis diagnosis function
- Built-in rotating platform, no need WR-3D stand, and save space



Ceiling-Mounted Dynamic DR

- Digital radiography and digital fluoroscopy
- Built in pneumoconiosis diagnosis function

3D Medical Images⁺



2D Medical Images⁺

